Aditya Gambhir

Software Engineer

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Summary

Software Engineer with 3+ years building production-grade microservices and event-driven backend systems processing enterprise-scale transaction volumes. Reduced release cycles from weeks to days through disciplined CI/CD, rigorous test-driven engineering, and 85% code coverage standards. Committed to bridging generative AI capabilities with distributed architecture to deliver secure, reliable, and high-.

Skills

Methodologies: SDLC, Agile (Scrum), Microservices Architecture, Waterfall
Programming Languages: Python, Java, TypeScript, Bash, SQL
Frontend Technologies: React, Angular, JavaScript, HTML5, CSS3, Redux, Bootstrap, jQuery
Backend Technologies: Spring Boot, Spring MVC, Spring Security, Hibernate, Node.js, Express.js, FastAPI
APIs & Testing Tools: RESTful APIs, GraphQL, Swagger, API Design, JUnit, Mockito, Jest, Cypress
Database Technologies: MySQL, PostgreSQL, MongoDB, DynamoDB, Kafka
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Azure, Docker, Kubernetes, Terraform, CI/CD
Build Tools & Version Control: Maven, Git, GitHub, Jenkins, GitHub Actions
AI/LLM Development: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, OpenAI API, Azure OpenAI, AWS Bedrock, Vector Databases (Pinecone, Chroma), Embeddings, Hugging Face, Agentic AI Frameworks (CrewAI, AutoGen)

Education

Master of Science in Computational Data Science | University of California, Riverside — **March 2025**
Bachelor of Technology, Computer Engineering | Vishwakarma Institute of Information Technology, India — **May 2023**

Work Experience

Software Engineer | Healthcare Support, LLC, California — **March 2026 - Current**

- Rebuilt a 7-year-old medical scheduling platform on **.NET 10**, **ABP Commercial**, and **Angular 20**, deploying to a self-hosted **Ubuntu** server, producing 846 commits, 390 pull requests, and 17 ADRs total.
- Migrated the platform from shared-database row filtering to one physical database per clinic office, validating isolation with a test harness that disables tenant filtering and confirms cross-office records stay invisible.
- Built two transactional outboxes using **SHA-256** idempotency keys and atomic database-lease claiming, plus reconciliation jobs with per-office error isolation, ensuring notifications are never lost or duplicated under concurrent worker processes.
- Ran a five-method bake-off for medical-record segmentation against hand-labeled ground truth; sliding-window won on Document-F1, recall, and cost, rejecting four alternative methods including text-based **OCR** consistently using fewer API calls.

Software Engineer | Centene, California — **January 2025 – March 2026**

- Built an **AI-powered** chatbot using **Spring Boot, Azure AI**, and **LLM**-based knowledge search to automate member support inquiries, cutting agent response time by 22% and documentation search time by 35%.
- Developed **REST APIs** with **OAuth2** authentication for internal and external member-facing systems, maintaining strict HIPAA compliance and 99.9% uptime across critical production healthcare environments and backend services daily.
- Engineered to a **Kafka**-based event streaming pipeline syncing member eligibility and claims data in real time across multiple internal healthcare systems, processing over 100,000 events daily with zero data loss.
- Wrote unit and integration tests using **JUnit** and **Mockito** for **backend services**, maintaining approximately 85% code coverage and catching defects before they reached the production release pipeline each week.
- Collaborated in **Agile Scrum** teams with Product, QA, and DevOps engineers, delivering new features in two-week sprints while conducting peer code reviews to maintain team engineering standards throughout each release.

Software Engineer | CRED, India — **January 2022 – August 2023**

- Developed 15+ **REST APIs** using **Java** and **Spring Boot**, writing automated **JUnit** tests to catch bugs early and reduce production issues before code changes reached the live payment processing system.
- Optimized slow-running **SQL** queries by rewriting joins and adding database indexes across services, cutting payment processing time by 40% for a system handling over 500,000 financial transactions each day.
- Configured **AWS** auto-scaling and access controls for **backend services** supporting real-time transaction processing, improving system uptime to 99.5% while reducing overall monthly cloud hosting infrastructure costs by 25%.
- Refactored key **ReactJS** components by introducing **Redux** state management and removing unnecessary repeated **API** calls, cutting page load time for 50,000 monthly active users on the app.
- Built 2 deployable **microservices** with automated unit tests in **Mockito** and created automated deployment pipelines using **Docker**, **Jenkins**, and **Kubernetes**, cutting release time from 2 weeks to 3 days.

Projects

Navigate LA 28- Real-Time Geospatial Navigation Platform (Project Link):

- Built a full-stack geospatial navigation platform using React 18/Redux, FastAPI, and PostgreSQL/PostGIS, delivering real-time interactive mapping, multi-modal route planning, and live transit tracking for Olympic venue navigation.
- Engineered a big-data analytics pipeline with Apache Spark and Hadoop/HDFS processing 1M+ records/minute, containerized the system via Docker Compose and Nginx, and hardened security (JWT, CORS, Bandit/Safety scans) as part of a 4-person team.

Survey Studio- AI-Assisted Literature Review System (Project Link):

- Architected a multi-agent literature review system using Microsoft AutoGen, orchestrating a Search Agent and Summarizer Agent to automate arXiv paper retrieval, analysis, and structured review generation via a FastAPI/Pydantic REST API.
- Built an LLM provider abstraction layer with intelligent fallback and cost optimization across Together AI, Gemini, Perplexity, and OpenAI, enforcing production-grade quality via Poetry, Ruff, Pyright, and CI, cutting manual literature review effort.